

rcmalli /
keras-vggface

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keras-vggface / keras_vggface / utils.py

...



iamgroot42 Update utils.py

6 years ago



97 lines (78 loc) · 3.74 KB

```
1  '''VGGFace models for Keras.
2
3  # Notes:
4  - Utility functions are modified versions of Keras functions [Keras](https://keras.io)
5
6  ...
7
8
9
10 import numpy as np
11 from keras import backend as K
12 from keras.utils.data_utils import get_file
13
14 V1_LABELS_PATH = 'https://github.com/rcmalli/keras-vggface/releases/download/v2.0/rcmalli_vggface_labels.txt'
15 V2_LABELS_PATH = 'https://github.com/rcmalli/keras-vggface/releases/download/v2.0/rcmalli_vggface_labels.txt'
16
17 VGG16_WEIGHTS_PATH = 'https://github.com/rcmalli/keras-vggface/releases/download/v2.0/rcmalli_vggface_weights.h5'
18 VGG16_WEIGHTS_PATH_NO_TOP = 'https://github.com/rcmalli/keras-vggface/releases/download/v2.0/rcmalli_vggface_weights_no_top.h5'
19
20
21 RESNET50_WEIGHTS_PATH = 'https://github.com/rcmalli/keras-vggface/releases/download/v2.0/rcmalli_resnet50_weights.h5'
22 RESNET50_WEIGHTS_PATH_NO_TOP = 'https://github.com/rcmalli/keras-vggface/releases/download/v2.0/rcmalli_resnet50_weights_no_top.h5'
23
24 SENET50_WEIGHTS_PATH = 'https://github.com/rcmalli/keras-vggface/releases/download/v2.0/rcmalli_senet50_weights.h5'
25 SENET50_WEIGHTS_PATH_NO_TOP = 'https://github.com/rcmalli/keras-vggface/releases/download/v2.0/rcmalli_senet50_weights_no_top.h5'
26
27
28 VGGFACE_DIR = 'models/vggface'
29
30
31 def preprocess_input(x, data_format=None, version=1):
32     x_temp = np.copy(x)
33     if data_format is None:
34         data_format = K.image_data_format()
35     assert data_format in {'channels_last', 'channels_first'}
36
37     if version == 1:
38         if data_format == 'channels_first':
39             x_temp = x_temp[:, ::-1, ...]
40             x_temp[:, 0, :, :] -= 93.5940
41             x_temp[:, 1, :, :] -= 104.7624
42             x_temp[:, 2, :, :] -= 129.1863
```

```

43         else:
44             x_temp = x_temp[..., ::-1]
45             x_temp[..., 0] -= 93.5940
46             x_temp[..., 1] -= 104.7624
47             x_temp[..., 2] -= 129.1863
48
49     elif version == 2:
50         if data_format == 'channels_first':
51             x_temp = x_temp[:, ::-1, ...]
52             x_temp[:, 0, :, :] -= 91.4953
53             x_temp[:, 1, :, :] -= 103.8827
54             x_temp[:, 2, :, :] -= 131.0912
55         else:
56             x_temp = x_temp[..., ::-1]
57             x_temp[..., 0] -= 91.4953
58             x_temp[..., 1] -= 103.8827
59             x_temp[..., 2] -= 131.0912
60     else:
61         raise NotImplementedError

```

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Code

Blame

Raw



```

66 def decode_predictions(preds, top=5):
67     LABELS = None
68     if len(preds.shape) == 2:
69         if preds.shape[1] == 2622:
70             fpath = get_file('rcmalli_vggface_labels_v1.npy',
71                             V1_LABELS_PATH,
72                             cache_subdir=VGGFACE_DIR)
73             LABELS = np.load(fpath)
74         elif preds.shape[1] == 8631:
75             fpath = get_file('rcmalli_vggface_labels_v2.npy',
76                             V2_LABELS_PATH,
77                             cache_subdir=VGGFACE_DIR)
78             LABELS = np.load(fpath)
79         else:
80             raise ValueError("`decode_predictions` expects '
81                             'a batch of predictions '
82                             '(i.e. a 2D array of shape (samples, 2622)) for V1 or '
83                             '(samples, 8631) for V2.'
84                             'Found array with shape: ' + str(preds.shape))
85     else:
86         raise ValueError("`decode_predictions` expects '
87                             'a batch of predictions '
88                             '(i.e. a 2D array of shape (samples, 2622)) for V1 or '
89                             '(samples, 8631) for V2.'
90                             'Found array with shape: ' + str(preds.shape))
91     results = []
92     for pred in preds:
93         top_indices = pred.argsort()[-top:][::-1]
94         result = [[str(LABELS[i].encode('utf8')), pred[i]] for i in top_indices]
95         result.sort(key=lambda x: x[1], reverse=True)
96         results.append(result)
97     return results

```

